

8	(a)	(i)	mean/average squared speed of the gas molecules				B1			
		(ii)	$\rho = \frac{\text{number of molecules } N \times \text{mass of one molecule } m}{\text{volume (of gas) } V}$ $pV = NkT$ or $pV = nRT$ correct algebra $p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle \Rightarrow pV = \frac{1}{3} Nm \langle c^2 \rangle \Rightarrow NkT = \frac{1}{3} \times \frac{2}{2} Nm \langle c^2 \rangle$ to obtain $\frac{1}{2} m \langle c^2 \rangle = \frac{3}{2} kT$ <i>Deduct 1 mark if symbols used are not defined but are conventional (e.g. V for volume, m for mass)</i> <i>Award zero if symbols used are not defined and unconventional.</i>				B1	B1	B1	
	(b)	internal energy = heat + work done direction of changes					B1 B1			
	(c)									
		(i)	no heat enters/leaves (gas)/no heating (of gas) due to “sudden expansion” negative work done / work done against atmosphere by expanding gas internal energy decreases					B1 M1 A1		
		(ii)	either volume decrease during melting, work done on ice/water or change in volume is small, negligible work done on ice heating of ice to break bonds internal energy increases					B1 M1 A1		
	(d)									
		(i)	Change in volume is zero hence zero work done, $Q = \Delta U$ $\Delta U = N_A E_k (= N_A \times \frac{3}{2} k \times 1)$ (because ideal gas, E_p is zero) Substitution of (1) $R(1) = N_A k(1)$ so, $Q = \frac{3}{2} R$					B1 B1 B1		
		(ii)	$q = mc\Delta T$ $c = \frac{\frac{3}{2} \times 8.31}{0.028 \times 1}$ $c = 450 \text{ J kg}^{-1} \text{ K}^{-1}$					C1 A1		
	(e)	change	heating supplied to gas / J	work done on gas / J	increase in internal energy / J					
		A → B	0	4.2	4.2					
		B → C	− 8.5	0	− 8.5					
			5.8	− 1.5	4.3					

		C → A	
Q	(c)	A nucleon is a proton or neutron. [D1]	